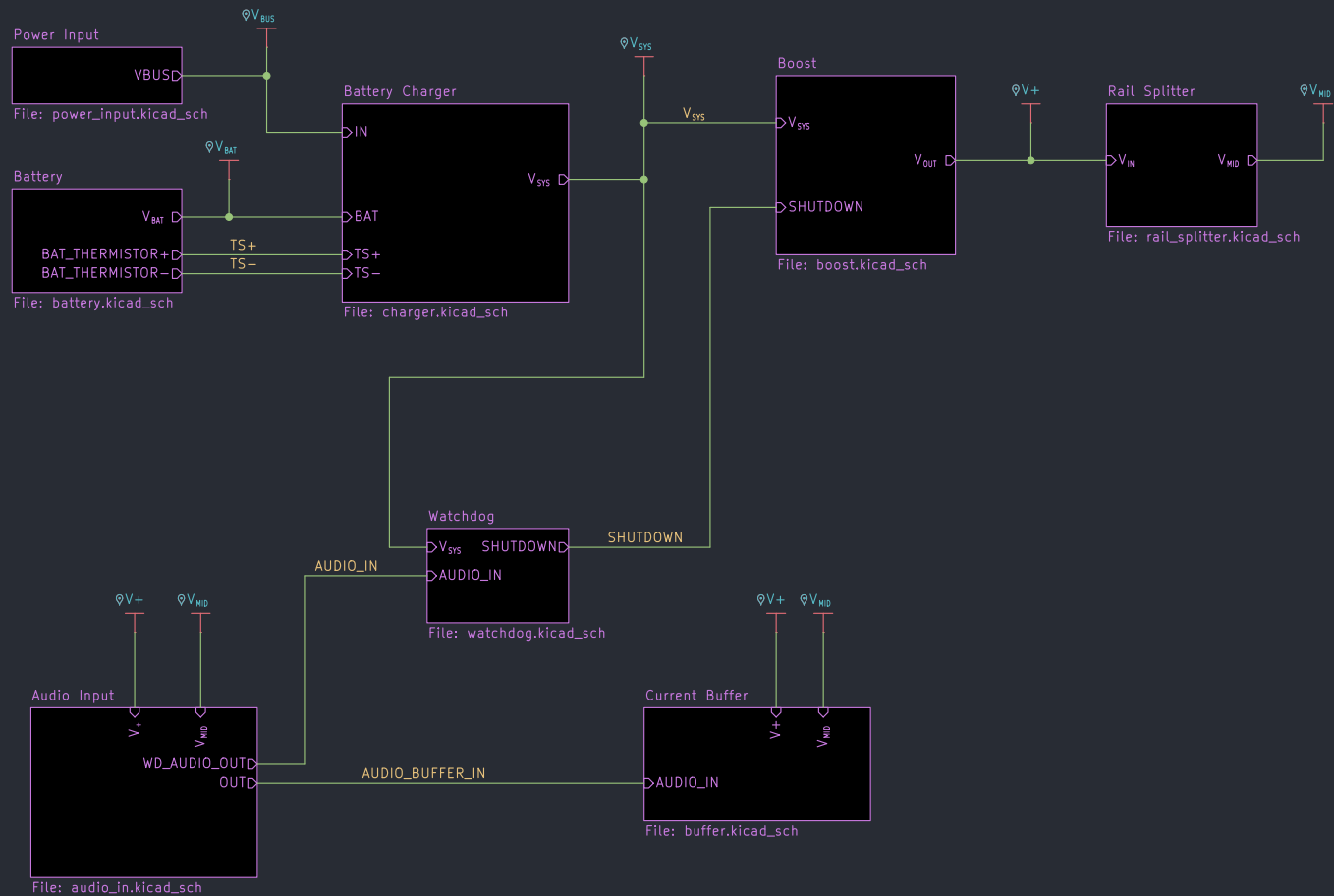




004-0001

Sheet: /
File: headphone_amp.kicad_sch

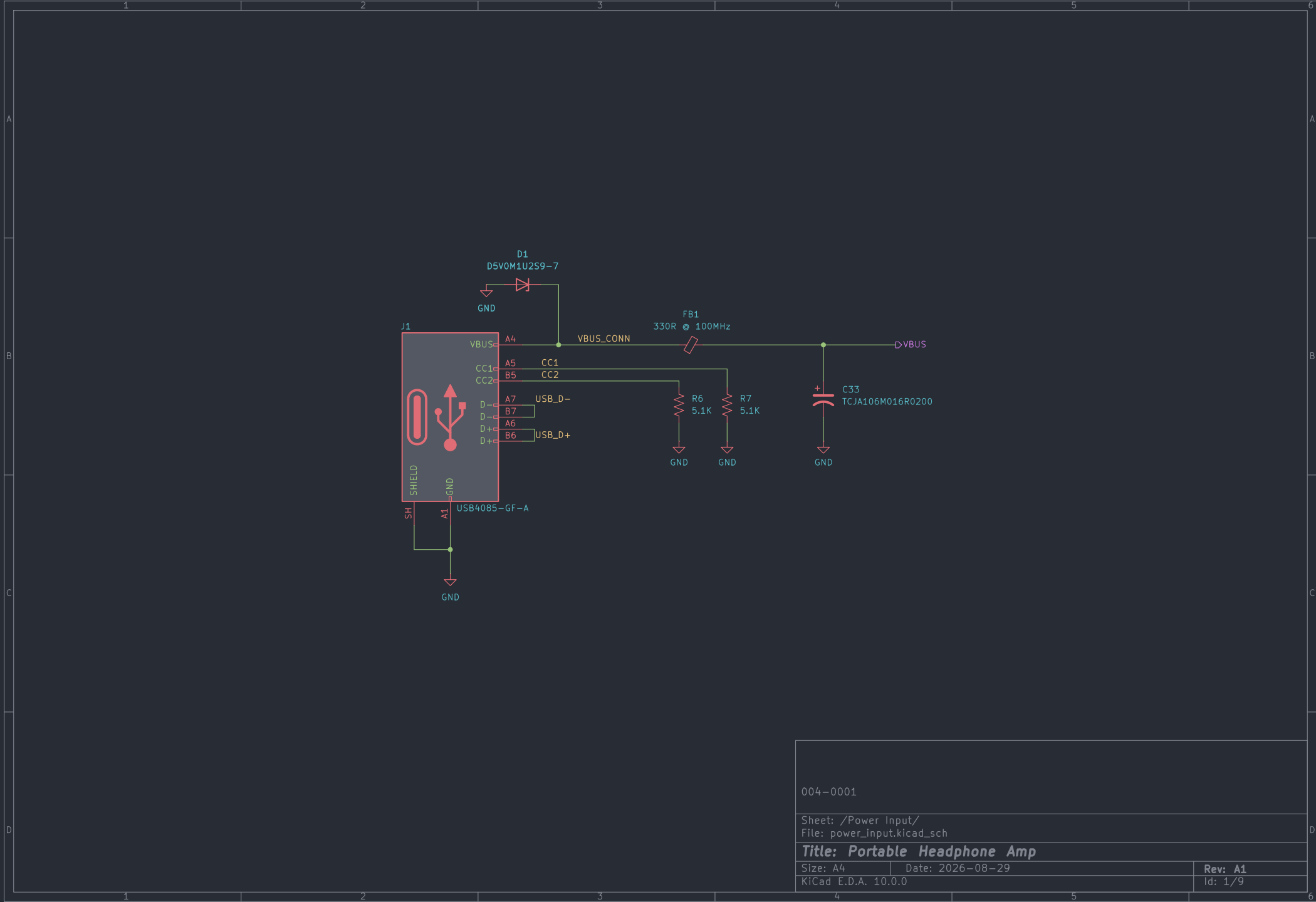
Title: Portable Headphone Amp

Size: A4 Date: 2026-08-29

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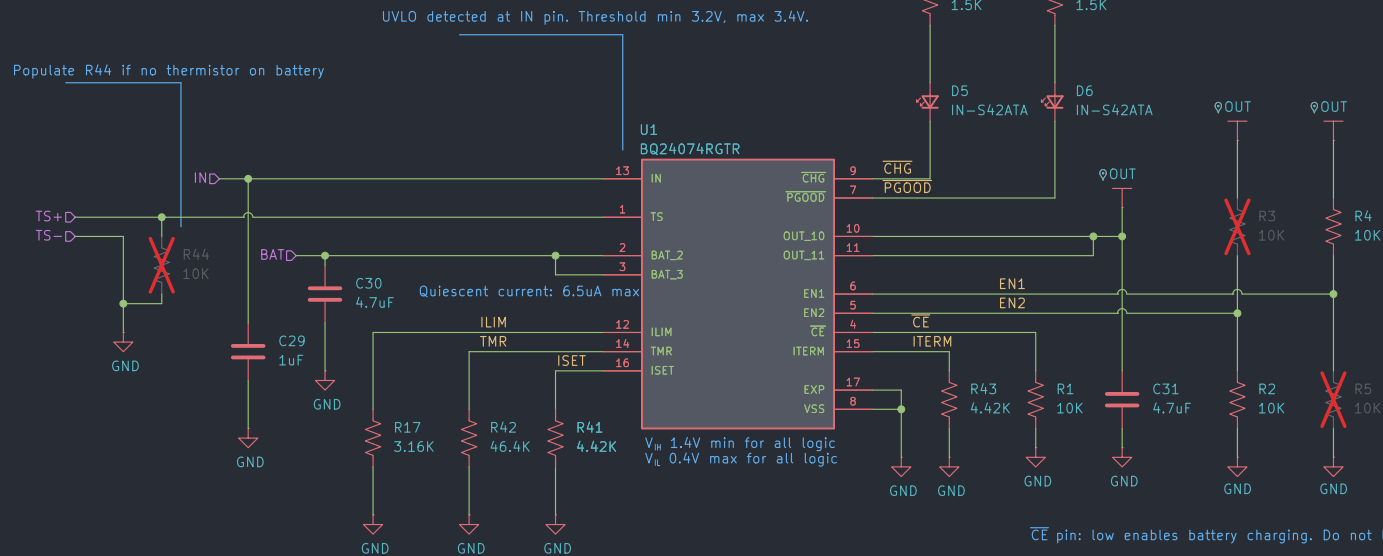


Table 7-2. EN1/EN2 Settings

EN2	EN1	MAXIMUM INPUT CURRENT INTO IN PIN
0	0	100 mA. USB100 mode
0	1	500 mA. USB500 mode
1	0	Set by an external resistor from ILIM to VSS
1	1	Standby (USB suspend mode)

$$I_{L_MAX} = K_{ILIM} / R_{ILIM}$$

$$K_{ILIM} = 1550 \text{ A}\cdot\text{Ohms}$$

$$\Rightarrow I_{L_MAX} = 1550 \text{ A}\cdot\text{Ohms} / 3.16\text{K} = 490 \text{ mA}$$

$$I_{CHG} = K_{ISET} / R_{ISET}$$

$$K_{ISET} = 890 \text{ A}\cdot\text{Ohms}$$

$$\Rightarrow I_{CHG} = 890 \text{ A}\cdot\text{Ohms} / 4.42\text{K} = 201 \text{ mA}$$

$$t_{MAXCHG} = R_{TMR} * 10 * K_{TMR}$$

$$K_{TMR} = 48 \text{ s}\cdot\text{kOhms}$$

$$\Rightarrow t_{MAXCHG} = 46.4\text{K} * 48 \text{ s}\cdot\text{kOhms} * 10 = 6.2 \text{ hours}$$

$$I_{TERM} = 0.030 * R_{ITERM} / R_{ISET}$$

$$R_{ISET} = 4.42\text{K}$$

$$\Rightarrow I_{TERM} = 0.030 * 4.42\text{K} / 4.42\text{K} = 30 \text{ mA}$$

004-0001

Sheet: /Battery Charger/
File: charger.kicad_sch

Title: Portable Headphone Amp

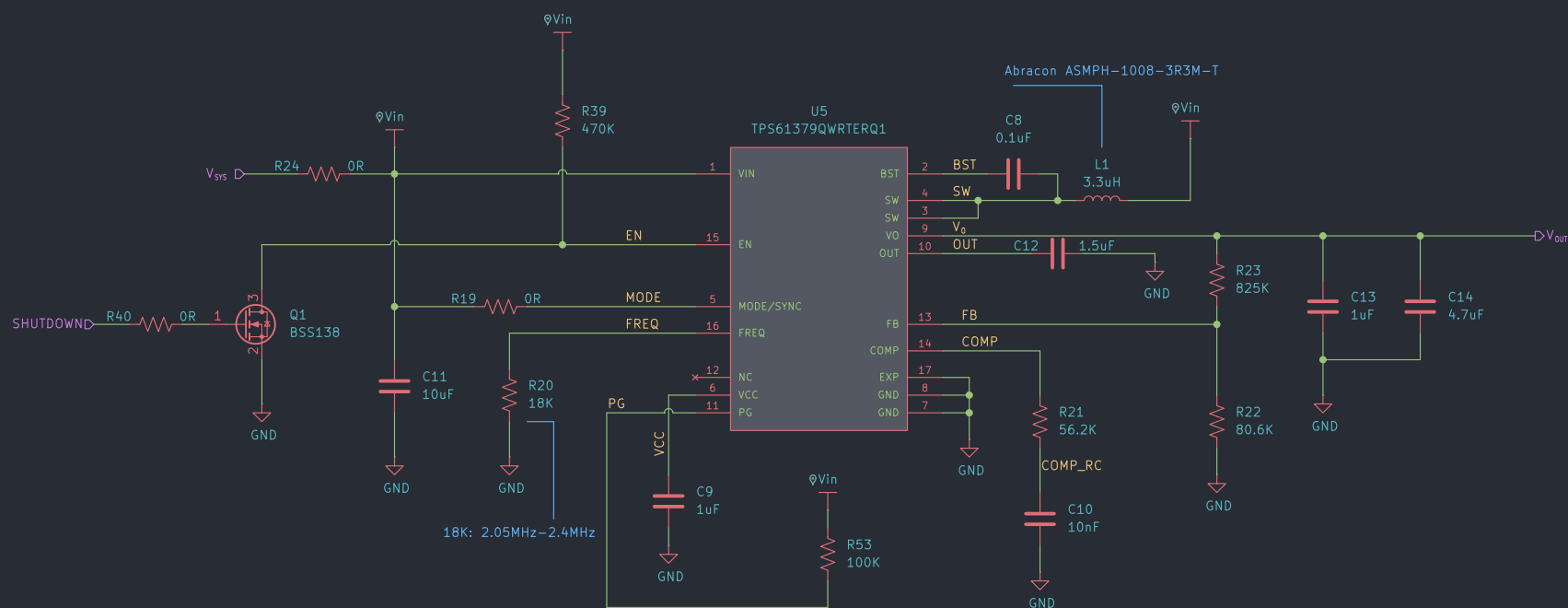
Size: A4 Date: 2026-08-29

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WEBENCH Design (PDF Link)

9V, 0.2A out



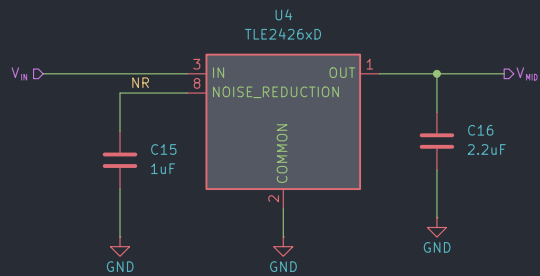
004-0001

Sheet: /Boost/
File: boost.kicad_sch

Title: Portable Headphone Amp

Size: A4 Date: 2026-08-29
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004-0001

Sheet: /Rail Splitter/
File: rail_splitter.kicad_sch

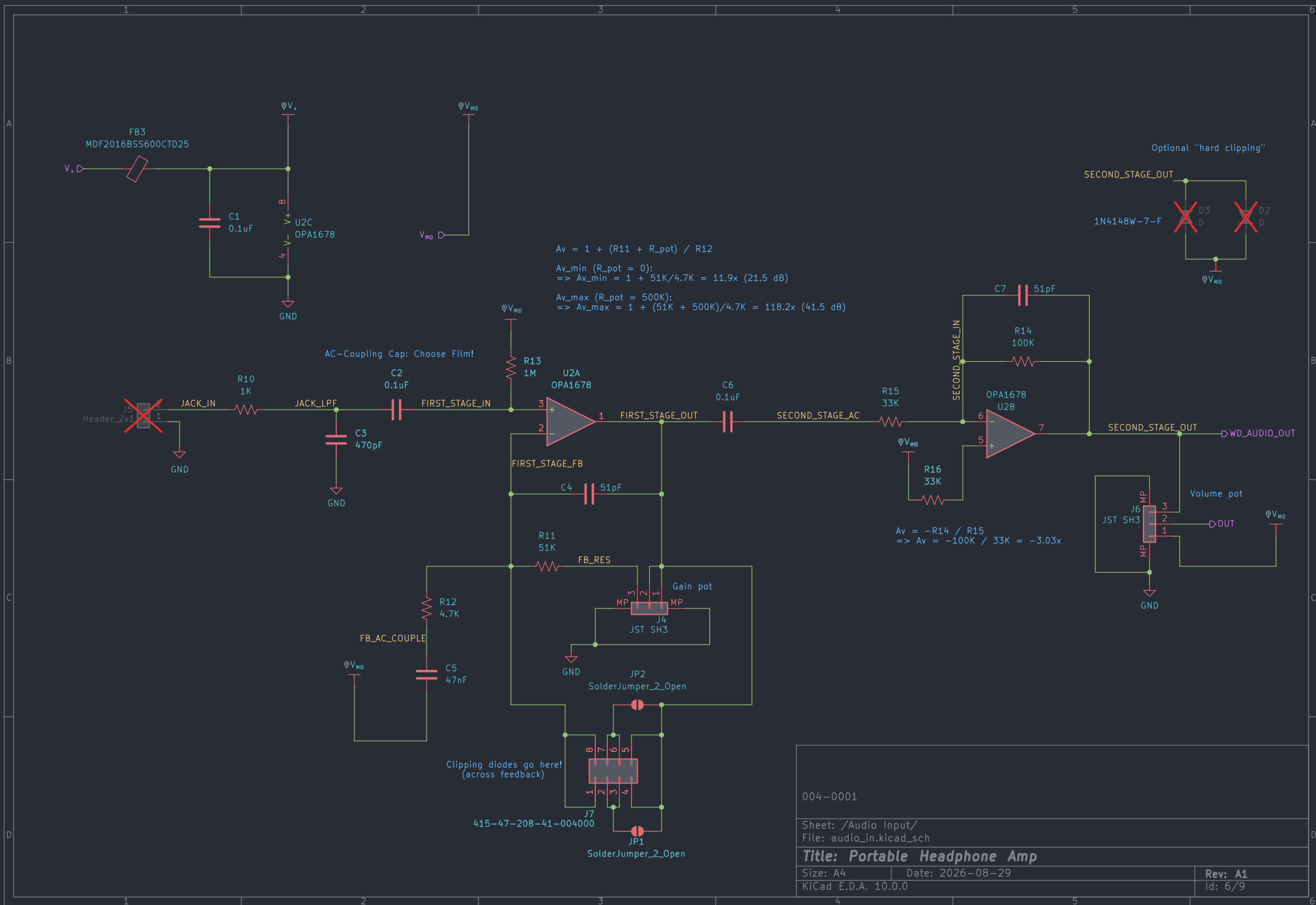
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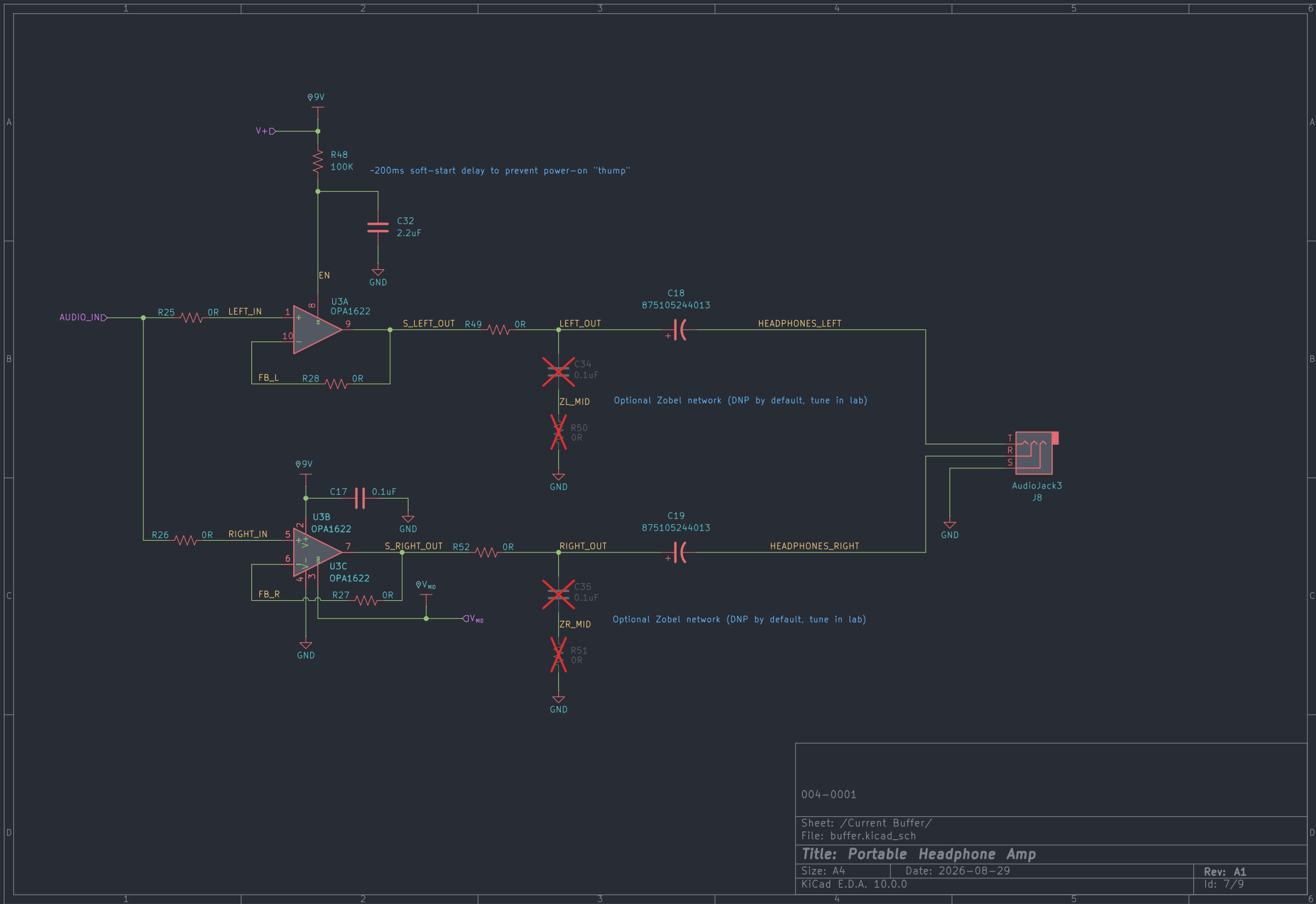
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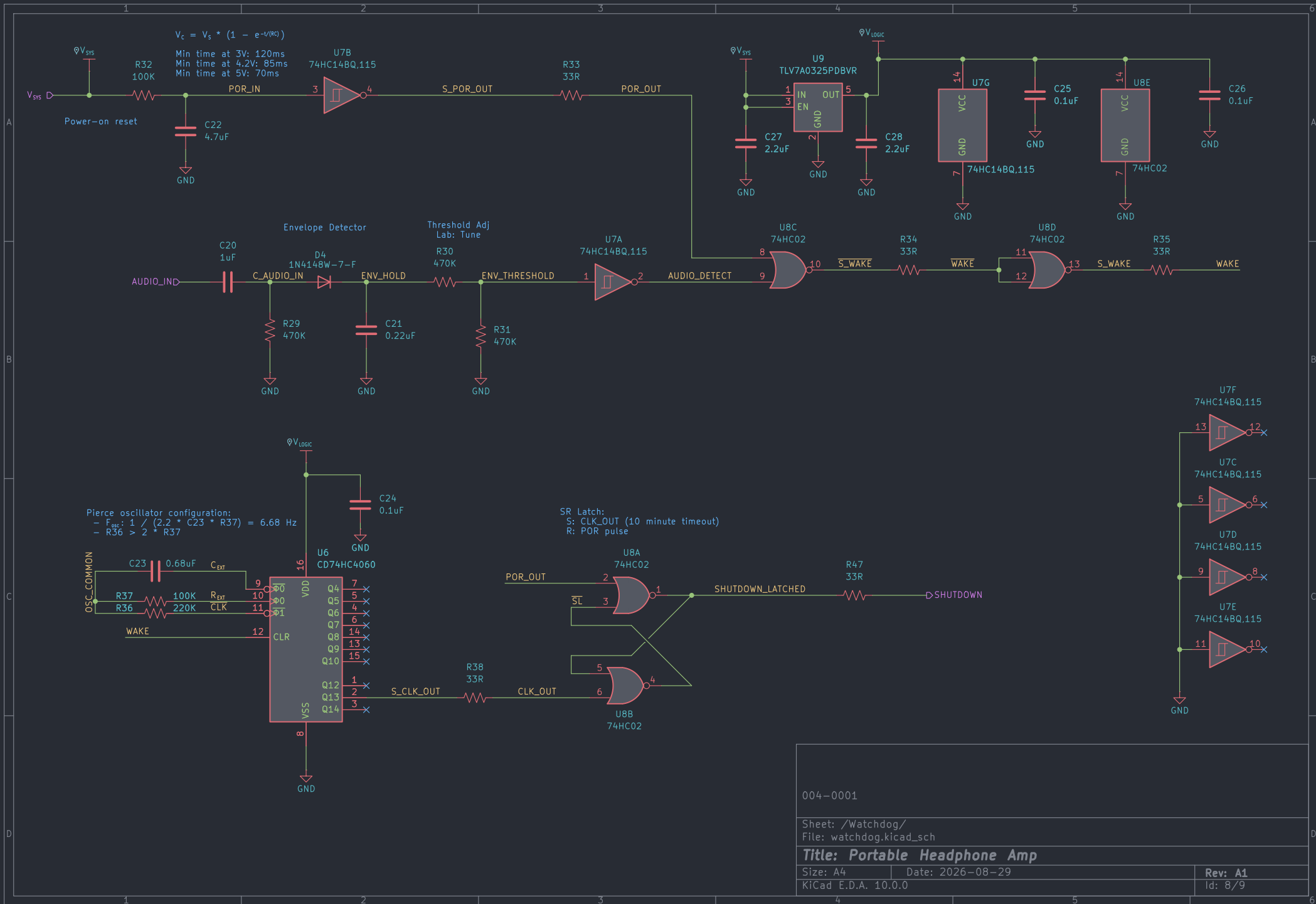
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004-0001

Sheet: /Watchdog/
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Title: Portable Headphone Amp

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